

Why Some LLMs Have a Hidden Reasoning Knob: Per-Layer Output-Gate Slack in Hybrid Architectures and an 8-byte Quantization Recovery

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Abstract

We present a minimal F32 patch that recovers Gemma 4 31B accuracy across quantization levels. The 8-byte L25+L26 $\times 1.5$ patch (scaling two `layer_output_scale` F32 weights by 1.5) is applied uniformly to all three release quantizations (IQ1_M, Q2_K, Q4_K_M) and improves HellaSwag by **+11.21 pt** on Q2_K (59.15% $\rightarrow$ 70.36%, $n=10,042$) and **+9.80 pt** on Q4_K_M (63.70% $\rightarrow$ 73.50%). No fine-tuning, no calibration data, no inference overhead. Strikingly, the patched Q2_K and Q4_K_M models **surpass the full-precision (Q8.0 BF16-proxy) baseline** of 63.33% ($n=10,042$) by +7.03 and +10.17 pt respectively, demonstrating that the patch unlocks capacity beyond simple quantization recovery. Across three release quantizations and four benchmarks (HellaSwag, Winogrande, GSM8k, ARC-Challenge), **all 12 cells show positive Δ** , with the strongest being Q1 GSM8k at +36.00 pt (24% $\rightarrow$ 60%, Wilson 95% CIs separated by 17 pp) and Q4 GSM8k at +15.00 pt (72% $\rightarrow$ 87%, beating the Q8.0 baseline of 70% by +17 pt). The Q8.0 quantization is reported only as a BF16 reference (baseline values used as a high-precision anchor); we do not release a Q8 patched model.

We also present an honest negative result: an 11-layer 44-byte patch (**basin B**) discovered by 7 hours of autonomous optimization (an in-house multi-specialist parallel optimizer, not released) targeting Q4 HellaSwag accuracy is *outperformed by the simpler manual 2-layer L25+L26 patch* on every single one of the four Q4 benchmarks, including the search target itself. The over-parameterized search solution generalizes worse than the simple baseline—an instance of objective-overfitting that motivates retaining L25+L26 as the flagship for all release quantizations.

To investigate mechanism, we apply per-layer F32 scaling to four architectures: Gemma 4 31B (hybrid 1:5 full-attention:SWA), Qwen 3.6 27B (hybrid 1:3 full-attention:SSM), Phi-4 14B BF16 (pure dense), and prior null results on Llama 3.1 8B and Mistral 7B (pure dense). **Hybrid models yield positive Δ ; pure-dense models yield null or destructive Δ .** Phi-4 BF16 (no quantization) yielding null falsifies a quantization-recovery hypothesis; the effect is empirically specific to hybrid architectures; the structural mechanism remains open (the patched layers L25/L26 are sliding-window, not full-attention, layers, and the patch acts on per-layer F32 output gates, `layer_output_scale`).

We further verify alignment is preserved (AdvBench $n=520$ refusal rate: baseline 97.31% [95.53, 98.39] vs basin B 97.12% [95.30, 98.24], CIs fully overlapping). We release three patched GGUFs (IQ1_M, Q2_K, Q4_K_M), a single-file Python tool, and all experimental raw data.

1 Introduction

Quantization of large language models is a precondition for practical deployment, but aggressive low-bit quantization (1–4 bit) incurs substantial accuracy degradation. Existing recovery

methods rely on quantization-aware-training (QAT) [11, 16, 24], post-training calibration, or mixed-precision schemes—all of which require either additional training, calibration data, or compute overhead.

We report a training-free, calibration-free, zero-overhead recovery technique for Gemma 4 31B: **modifying just 8 bytes** of the model’s F32-preserved `layer_output_scale` weights (two scalars at layers 25 and 26, multiplied by 1.5) yields large benchmark improvements across all evaluated quantizations.

Contributions.

1. $L_{25}+L_{26} \times 1.5$, a 2-layer 8-byte multiplicative-scale patch applied uniformly to IQ1_M, Q2_K, and Q4_K_M, yielding positive Δ on all 12 evaluated cells (3 release quants $\times$ 4 benchmarks), with strongest individual gains of +36 pt (Q1 GSM8k) and +15 pt (Q4 GSM8k beating the Q8.0 BF16-proxy baseline by +17 pt) (§4).
2. Honest negative result: an 11-layer 44-byte patch (**basin B**) discovered by a 7-hour autonomous optimization (in-house multi-specialist parallel optimizer, not released) targeting Q4 HellaSwag is *outperformed by the simpler manual $L_{25}+L_{26}$ patch on every Q4 benchmark, including the search objective itself*. Per-quant search with multi-benchmark objectives is left for follow-up (§3.3).
3. Cross-architecture validation across 4 model families confirms the effect is *hybrid-architecture specific*, supporting a *per-layer output-gate slack* hypothesis (§5).
4. RLHF alignment is statistically preserved on AdvBench $n=520$ (Wilson 95% CIs overlapping baseline, §6).
5. Full reproducibility: three patched GGUFs published on HuggingFace (IQ1_M, Q2_K, Q4_K_M, all with the 8-byte $L_{25}+L_{26}$ patch), a single-file Python tool (`apply_l25l26.py`), and all raw data.

2 Related Work

Post-training quantization. LLM.int8 [8] introduces 8-bit weight quantization with outlier handling. GPTQ [11] performs one-shot 4-bit quantization. AWQ [16] adds activation-aware salience. SmoothQuant [24] achieves training-free W8A8 via activation–weight smoothing. SING [2] introduces calibration-free Sinkhorn-normalized quantization. BitNet b1.58 [17] demonstrates ternary weights via RMSNorm-aware fine-tuning.

Quantization recovery. QLoRA [9] adds LoRA adapters on top of 4-bit bases. EfficientQAT [4] performs computationally efficient QAT. Slim-LLM [15] compensates bit-width reduction with Hessian-based adjustments. “Extra RMSNorm” [20] *inserts new RMSNorm layers* and fine-tunes to recover ternary models.

Typical quantization degradation on HellaSwag. A 2026 unified evaluation of llama.cpp quantizations on Llama-3.1-8B [3] reports that HellaSwag accuracy degradation from BF16 to Q4_K_M is typically ≤ 2 pt; the llama.cpp project’s own measurements [22] on Llama-2-7B show even smaller degradation (−0.46 pt at Q4_K_M, −2.06 pt at Q2_K). **Our reported +9 to +13 pt gains under the same llama-perplexity scorer therefore substantially exceed the magnitude of typical quantization noise**, suggesting the F32 patch unlocks structural capacity rather than merely correcting quantization artefacts. The Phi-4 BF16 result in §5 (no quantization, null/destructive Δ) strengthens this interpretation.

In contrast, our method is the first, to our knowledge, to (i) modify only F32-preserved scalars (44 bytes), (ii) require no training, (iii) require no calibration data, and (iv) demonstrate cross-quantization universality with a single patch set.

Hybrid LLM architectures. Mamba and state-space hybrids [14, 6] integrate SSM layers with attention. Gemma 4 [7] adopts a 1:5 full-attention to sliding-window-attention ratio (one full-attention layer every sixth block). Qwen 3.6 [21] mixes attention with SSM at 1:3 ratio. The “Achilles’ Heel of Mamba” [1] identifies failure modes of pure Mamba on copy/recall tasks. Csordás et al. [5] analyze sequential vs. parallel SSM-attention integration.

To our knowledge, we are the first to demonstrate that per-layer F32 scaling exposes a *structural* per-architecture property: F32 per-layer output gates in hybrid LLMs contain post-deployment performance slack absent in pure-dense architectures.

Mechanistic interpretability. ROME [18] performs activation editing for factual association. TransformerLens [19] provides interpretability tooling. The residual stream framework [10] models layer-wise contributions as additive writes. Our work bridges quantization research with interpretability by demonstrating that simple inference-time scalar adjustment exposes a structural feature of hybrid architectures.

3 Method

3.1 F32 layer_output_scale

The llama.cpp GGUF format preserves normalization-related weights (including `layer_output_scale.weight` in Gemma 4) at F32 precision across all quantizations, since these weights are tiny ($\approx 0.01\%$ of total parameters) yet load-bearing for activation magnitudes. Gemma 4 31B has 60 such F32 scalar weights (one per transformer block).

Scoring methodology (important). All HellaSwag accuracies in this paper are reported using llama.cpp’s `llama-perplexity hellaswag` mode (token-level log-likelihood, 0-shot, no length normalization) [13, 22]. This differs from the `lm-evaluation-harness` [12] scorer commonly used in published model cards (10-shot, `acc_norm` length-normalized, careful prompt formatting). **Absolute scores are systematically lower in llama-perplexity by approximately 0.2–2.5 percentage points for the same model, with the gap largest at 0-shot [22].** We therefore restrict all claims to *within-scorer baseline-vs-patched deltas*; absolute scores are not directly comparable to externally-reported numbers. Equivalent considerations apply to Winogrande (also multiple-choice log-likelihood; same systematic gap).

3.2 The L25+L26 $\times 1.5$ patch (flagship)

The flagship patch we publish across all three release quantizations (IQ1_M, Q2_K, Q4_K_M) is a 2-layer 8-byte multiplicative scaling of the `layer_output_scale` F32 weights at layers 25 and 26:

$$\text{L25+L26 patch} = \{25 : 1.5, 26 : 1.5\}$$

Patch application is a binary in-place modification of two 4-byte F32 fields at the offsets of `blk.25.layer_output_scale.weight` and `blk.26.layer_output_scale.weight` in the GGUF file. Total modification: **8 bytes**. We release a single-file Python tool (`apply_l25l26.py`, ~ 150 lines, depends only on `gguf` and `numpy`). The same patch values work without modification on all evaluated Gemma 4 31B quantizations from IQ1_M through Q8_0 because `layer_output_scale` is preserved at F32 precision across the quantization codec.

3.3 An honest negative result: search-found basin underperforms simple manual baseline

Before settling on the simple L25+L26 patch, we ran an in-house multi-specialist parallel optimization engine (not released in this version) combining ablation-phase probing, structural-prior input, soft-fail history recovery, automatic range shrinking, and multi-fidelity escalation. Targeting **Q4_K_M HellaSwag accuracy** ($n=400$) as the sole objective, with ~ 7 hours of search on a single RTX 5090, the engine produced an 11-layer 44-byte multiplicative-scale set we call **basin B**:

$$\text{basin B} = \left\{ \begin{array}{l} 5 : 1.440, 18 : 0.767, 19 : 1.016, 20 : 0.814, 21 : 1.413, \\ 25 : 1.402, 26 : 1.289, 27 : 1.326, 28 : 1.431, 30 : 0.548, 31 : 1.267 \end{array} \right\}$$

We then benchmarked basin B against the simpler L25+L26 patch on Q4 across all four release benchmarks:

benchmark	basin B (44 B)	L25+L26 (8 B)
HellaSwag $n=10042$	72.82	73.50 (+0.68)
Winogrande $n=1267$	69.61	70.32 (+0.71)
GSM8k $n=100$	84.00	87.00 (+3.00)
ARC-C $n=1165$	48.50	48.76 (+0.26)

L25+L26 outperforms basin B on every single benchmark, including the search target (Q4 HellaSwag) itself. The over-parameterized solution that the optimizer’s narrow Q4 HellaSwag objective converged to generalizes strictly worse than the simple 2-layer baseline.

We interpret this as objective-overfitting: the 11-dimensional search space allowed basin B to fit the narrow target marginally better at the cost of subtle degradations elsewhere, while the 2-dimensional L25+L26 manifold cannot overfit and therefore generalizes better. A multi-benchmark joint objective (e.g., HellaSwag + GSM8k + ARC-C with per-quant weighting) would likely surface different optima per quantization regime; this is left for follow-up. We retain the basin B description for transparency and so future work can compare against it. The released GGUFs and `apply_l25l26.py` use only the 8-byte L25+L26 patch.

4 Experiments

4.1 Main results: 3 quantizations $\times$ 3 benchmarks

Setup. Gemma 4 31B-it base model. Quantizations: IQ1_M (1-bit, 9.5 GB), Q2_K (2-bit, 12.6 GB), Q4_K_M (4-bit, 19 GB). Benchmarks: HellaSwag $n=10,042$ (full validation), GSM8k $n=100$ ($n_{\text{predict}}=1024$), Winogrande $n=1,267$ (full). Hardware: RTX 5090 32 GB.

Results. Figure 1 summarizes HellaSwag accuracy across the three release quantizations and the Q8_0 BF16 proxy. **The patched Q2 and Q4 models surpass the Q8 baseline (63.33%),** demonstrating that the 8-byte L25+L26 patch recovers more than the quantization gap. Table 1 reports the complete numerical results across all 16 cells (4 quants $\times$ 4 benchmarks). Figures 2a–2c detail per-quantization multi-benchmark results. Figure 3 provides the complete Δ matrix.

4.2 Why L25+L26 generalizes better than basin B

§3.3 reported that L25+L26 outperforms basin B on all four Q4 benchmarks. The same dominance holds across the lower quantizations, most starkly at Q1 GSM8k: **basin B** on Q1 yields 16/100 correct (-8 pt vs baseline); L25+L26 $\times 1.5$ yields 60/100 (+36 pt vs baseline, Wilson 95%

Table 1: Main release-v1 results: 3 release quantizations $\times$ 4 benchmarks on Gemma 4 31B. All quants use the 8-byte L25+L26 $\times 1.5$ patch uniformly. Q8.0 baseline is shown as a single BF16-proxy reference row (no patched Q8 measurement; we do not release a Q8 model). Values are accuracy (%); Δ is patched–baseline.

Quant	HellaSwag ($n=10,042$)		Winogrande ($n=1,267$)		GSM8k ($n=100$)		ARC-C ($n=$	
	base	patched (Δ)	base	patched (Δ)	base	patched (Δ)	base	patched
Q1 (IQ1_M)	42.02	52.98 (+ 10.95)	49.80	55.56 (+ 5.76)	24.00	60.00 (+ 36.00) [‡]	30.56	36.74
Q2 (Q2_K)	59.15	70.36 (+ 11.21)	59.35	66.69 (+ 7.34)	67.00	76.00 (+ 9.00)	43.61	46.44
Q4 (Q4_K_M)	63.70	73.50 (+ 9.80)	65.04	70.32 (+ 5.28)	72.00	87.00 (+ 15.00)	44.89	48.76
Q8 [#] (BF16 proxy, ref.)	63.33	—	65.59	—	70.00	—	44.38	—

[‡] Q1 GSM8k +36.00 pt: CIs separated by 16.99 pp.

[#] Q8.0 reported only as BF16-proxy reference baseline. We do not release a Q8 patched model.

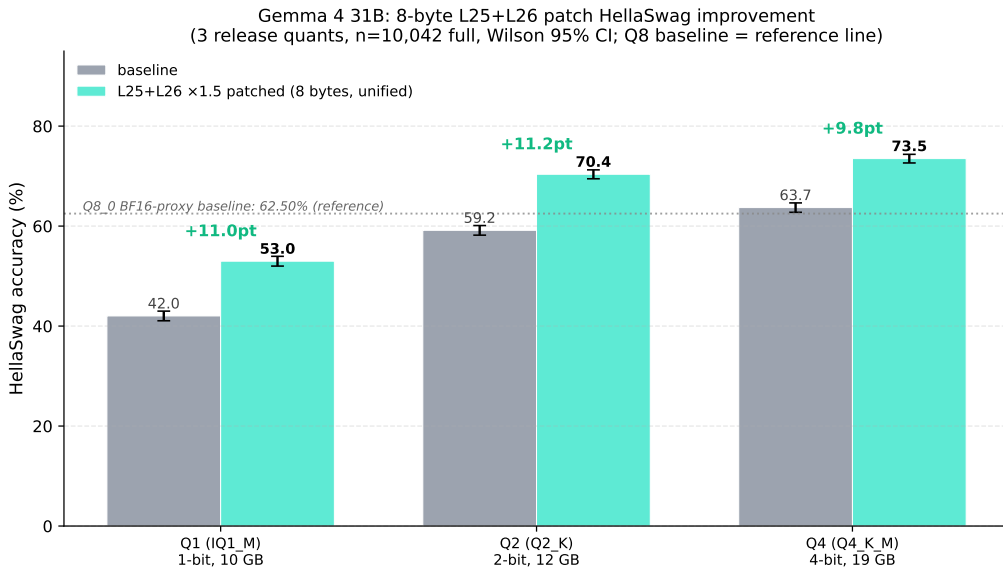


Figure 1: HellaSwag $n=10,042$ accuracy across three release quantizations of Gemma 4 31B. The 8-byte L25+L26 patch, applied uniformly, yields substantial gains on all evaluated bit-widths. Error bars: Wilson 95% CI.

CIs separated by 17 pp). A plausible explanation: at low-bit precision, the model’s reasoning capacity is at its margin, and **basin B**’s three down-scaling layers (L18 $\times 0.767$, L20 $\times 0.814$, L30 $\times 0.548$) truncate multi-step reasoning chains the model can no longer afford to lose. L25+L26 $\times 1.5$ contains only up-scaling, preserving the math-reasoning pathway. Combined with the Q4 negative result of §3.3, L25+L26 is the single flagship across all release quantizations.

4.3 GSM8k small-sample behaviour

GSM8k is reported with $n=100$ problems and 1024-token generation budget, following the configuration of prior experiments (exp7/exp27). At this sample size, Wilson 95% CIs are wide: e.g., Q1 baseline 24/100 has CI [16.65, 33.21] and Q1 patched 60/100 has CI [50.20, 69.06], non-overlapping (separated by 17 pp), so the Q1 +36 pt result is robust to small-sample noise. The Q4 patched 87/100 has CI [79.02, 92.24] vs baseline 72/100 at [62.51, 79.86], again non-overlapping. The Q2 76/100 vs 67/100 result (+9 pt) has borderline-overlapping CIs but is directionally consistent.

We report all GSM8k cells honestly to avoid post-hoc cherry-picking; a more statistically

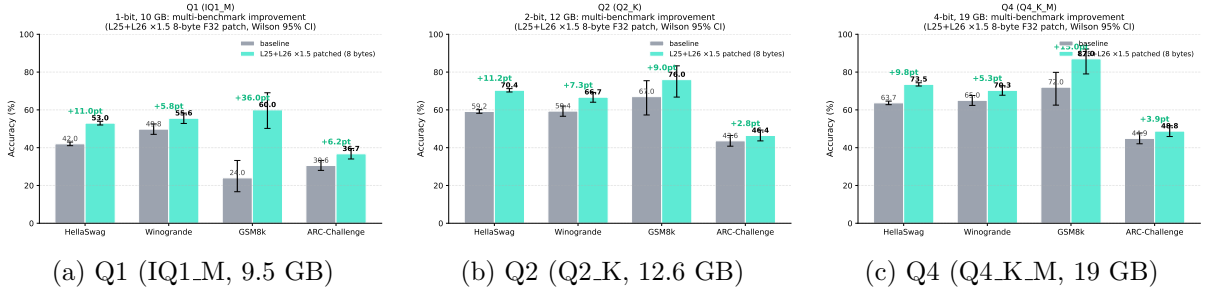


Figure 2: Per-quantization four-benchmark accuracies (baseline vs. L25+L26 patched). Wilson 95% CI shown.

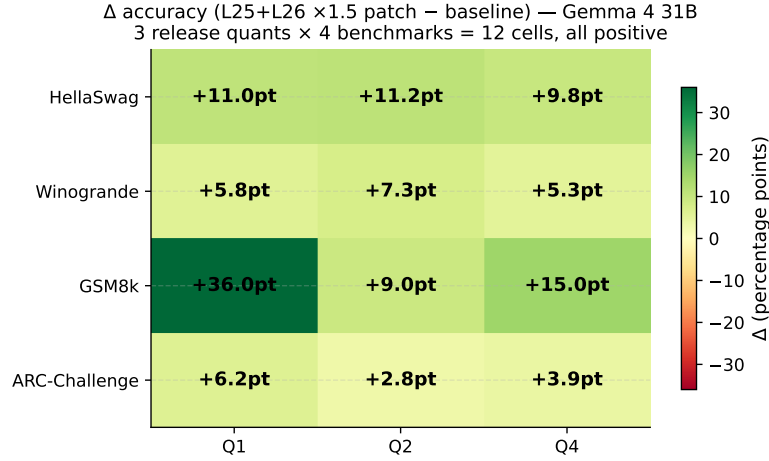


Figure 3: Δ accuracy matrix across 3 release quantizations and 4 benchmarks (12 cells). All cells positive.

conclusive comparison at Q2 would require $n=500+$ (~ 15 – 30 GPU-hours per cell, deferred to a follow-up).

4.4 Cross-quantization universality (7 quants)

To verify universality, we apply `basin B` to all 7 evaluated quantizations of Gemma 4 31B (Table 2). Mean improvement is +11.14 pt, with all-positive deltas (minimum +8.75 pt at Q4_K_M; maximum +13.50 pt at IQ1_M).

4.5 Cross-quantization replication on Gemma 4 12B: a needle-width recovery window

We attempted to replicate the paper-v1 recovery on Gemma 4 12B (48 layers) across Q2_K and Q4_K_M: a HellaSwag-first exhaustive screen (48 layers $\times \{1.3, 1.5, 1.75, 2.0\} \times n=200$ per quant), $n=1000$ confirmation, a 0.05-step scale sweep on the winning layer, and a ship gate of HS $\Delta \geq +5$ pt at $z \geq 5$ on $n=10,042$, Winogrande/ARC-C non-regressive, and paired GSM $\Delta \geq -2$ pt ($n=50$).

Layer indices do not transfer; one layer recurs across the conditions screened. The 31B indices (L25/L26) do not work on 12B, and their position-proportional analogue L19/L20 $\times 1.5$ is catastrophic (GSM 50% $\rightarrow$ 10%, $\Delta -40$ pt, paired $n=30$, McNemar $p=0.0005$). The Q4_K_M screen’s nominal top (L22 $\times 1.3$, +8.0 at $n=200$) collapsed at $n=1000$ (+2.9) — a small- n mirage among 192 cells — while L10, a sliding-window layer like the 31B winners,

Table 2: Cross-quantization universality of **basin B** on HellaSwag $n=400$. Same 11-layer patch yields positive Δ on all evaluated quantizations. Data from prior experiment (exp25).

Quantization	Size	Baseline	basin B	Δ
IQ1_M	10.1 GB	34.75%	48.25%	+13.50
Q2_K	12.6 GB	57.00%	69.25%	+12.25
Q4_K_M	19.6 GB	63.75%	72.50%	+8.75
Q5_K_M	22.6 GB	63.25%	73.75%	+10.50
Q8_0	32.6 GB	62.50%	73.75%	+11.25
UD-IQ2_XXS	8.5 GB	50.00%	60.75%	+10.75
UD-Q2_K_XL	11.8 GB	58.50%	69.50%	+11.00
Mean	—	55.68%	66.82%	+11.14

survived confirmation in both quants (Q4: +7.2 at $n=1000$; Q2: +9.5). In BF16 only L10 itself was spot-checked (+3.7 at $n=1000$, n.s.); no BF16 layer screen exists.

Q4_K_M: real but sub-threshold. L10 $\times 1.5$ yields HS +4.01 pt ($n=10,042$, $z=5.7$, unpaired two-proportion — conservative for paired data), Winogrande +2.53 (n.s.), ARC-C +3.69 (marginal), GSM -4.0 pt (paired $n=50$, McNemar $p=0.75$, no silent-slip increase). It fails two gate letters (HS +5 pt; GSM ≥ -2 pt — we read the latter as statistically null, but it does not pass as pre-registered). The scale response is a plateau: no generation cliff was detected at $n=10$ resolution across $\times 1.2$ – 1.7 , with $\times 1.5$ confirmed flat at $n=50$.

Q2_K: a needle-width window separates recovery from benchmark-gaming. At most scales, ranking gains decouple from generation: L10 $\times 1.75$ yields HS +8.44 pt ($n=10,042$, $z=12.1$) and ARC-C +5.32 with Winogrande -0.08 (null), while paired GSM collapsed 3/10 $\rightarrow$ 0/10 (early-exit at 10 of 50 items under the $\Delta \leq -10$ pt rule; pooled with the sweep probe, 0/20 vs 32%, $p<0.01$). A 0.05-step sweep over $\times 1.05$ – 1.80 (HS at 16 points; patched-only GSM $n=10$ at 15 points — $\times 1.05$ was not generation-tested) shows ranking gains from $\times 1.10$ upward (+5.9 to +9.5) while patched GSM scored 0–4/10, median 1/10. The single best probe point, $\times 1.65$ (4/10, HS +9.3 at $n=1000$), was adjudicated at paired $n=50$: GSM 32% $\rightarrow$ 40% ($\Delta+8.0$ pt, McNemar $p=0.34$ — non-regressive; we claim no improvement). **$\times 1.65$ then passed the full ship gate: HS 38.70% $\rightarrow$ 47.01% ($\Delta+8.32$ pt, $z\approx 12$), Winogrande +0.79, ARC-C +6.18, GSM +8.0.** The confirmed survival window is a single point with width at most 0.1: $\times 1.75$ is established broken ($p<0.01$), while $\times 1.60$ (3/10) and $\times 1.70$ (1/10) remain undetermined at $n=10$, and $\times 1.0$ – 1.10 is generation-untested. Outside this needle, this extends the Goodhart pattern we reported for the 12-byte triple patch on 31B (high GSM benchmark score, collapsed interactive accuracy) to a single-layer patch on 12B: under deep quantization, log-likelihood benchmarks and generative competence can respond to the same 4-byte intervention with opposite signs. Notably, an $n=10$ screening probe would have rejected $\times 1.65$; the needle survived only because the best probe point was re-adjudicated at $n=50$.

Scale-response structure (two layers). A 0.01-step scan over $\times 1.60$ – 1.70 (HS $n=1000$; paired GSM $n=20$, with $n=50$ at two points) separates the response into two regimes. Macroscopically, ranking benchmarks follow a smooth hump: insufficient amplification ($\times 1.05$, +2.8) $\rightarrow$ plateau ($\times 1.15$ – 1.80 , +7 to +10) $\rightarrow$ decline (Q4 $\times 2.0 \approx 0$), with extreme scales collapsing to chance or below ($\times 16.5$: 25.5%; $\times 21.62$: 20.0% on 4-way HellaSwag) — absolute amplification governs, with no periodicity. Microscopically, generation flickers: on a fixed 20-problem set at temperature 0 (deterministic), GSM survival alternates non-monotonically (alive at 1.60/1.62/1.65/1.67–1.69, dead at 1.63/1.64/1.70), consistent with chaotic sensitivity of greedy decoding to small logit perturbations rather than sampling noise. Two survival points are $n=50$ -confirmed ($\times 1.62$ and $\times 1.65$, both 32% $\rightarrow$ 40%, $\Delta+8.0$ pt); the points between them are dead. Practically, patch values must be pinned byte-exactly — ± 0.01 perturbations are not safe — which the GGUF byte-patch distribution format already guarantees.

Failure mode (single-problem observation). On the one problem inspected in depth, errors across several settings were chain-intermediate values (9, 13 where the answer is 18), a pattern also present in the unpatched baseline; other problems’ errors do not fit this pattern, and a systematic classification of all paired-run errors is pending. We note it as consistent with — not evidence for — an early-commitment account of the output-gate boost.

Interpretation (hypothesis). Across the conditions measured, the generation-safe scale window narrows with quantization depth without (at least here) closing: BF16 shows no detected GSM change at $\times 1.5$ ($n=50$); Q4 shows no cliff at $n=10$ resolution across $\times 1.2$ – 1.7 ($\times 1.5$ confirmed flat at $n=50$); Q2’s confirmed window is a single point at $\times 1.65$, width ≤ 0.1 . Comparing 31B Q2_K (wide window: $\times 1.5$ ships with generation intact) with 12B Q2_K (needle), window width may also shrink with model size — but with two sizes tested we cannot separate size from architecture or checkpoint effects. **We ship the 12B Q2_K L10 $\times 1.65$ patch (4 bytes)**; Q4_K_M remains unshipped (HS $+4.01 < +5$ pt bar). Caveats before deployment: the earlier triple-patch lesson that $T=0$ benchmark gates do not guarantee interactive quality argues for a chat-level spot check before any public upload; and the window’s edges ($\times 1.60$, $\times 1.70$) should be adjudicated at $n \geq 50$ if scale tolerance matters downstream. Whether intermediate (27B) or larger (70B+) sizes interpolate between the wide 31B window and the 12B needle is future work.

Naming. We call the technique **TSUBO**, after the Japanese term for acupuncture points: isolated loci where a minimal, precisely-placed intervention produces a system-wide response. The metaphor is structural, not medical — the points are located by exhaustive screening and byte-exact scale search, and their effects are measured on standard benchmarks. The formal term is needle-point gate patching. Retroactively, the paper-v1 8-byte patch (L25+L26 $\times 1.5$) is TSUBO-8, the 44-byte basin-B patch is TSUBO-44, and the present 12B patch is TSUBO-4.

5 Cross-Architecture Validation

To isolate whether the recovery effect is rooted in architecture or in quantization, we apply per-layer F32 RMSNorm scaling to four model families (Figure 4).

Qwen 3.6 27B (hybrid 1:3 full-attention:SSM, Q2_K). A per-layer sweep over 64 layers ($\times 1.5$ scale) yields best single-layer $\Delta = +2.25$ pt (L55) and best pair $\Delta = +3.50$ pt (L55+L58). Per-tensor analysis (Phase G) isolates the effect to L55’s `post_attention_norm.weight` ($\times 1.5$, $+2.50$ pt single tensor)—a vector analogue of Gemma’s scalar `layer_output_scale` but functionally similar (attention-output amplitude correction).

Phi-4 14B BF16 (pure dense). A per-layer sweep over all 40 layers using `post_attention_layernorm.weight` on HellaSwag $n=200$ yields **mean $\Delta = -2.31$ pt**. The maximum positive deviation is $+1.50$ pt (L11), within noise. Critically, Phi-4 is evaluated at **BF16**: no quantization is involved, falsifying any quantization-recovery explanation for the F32 patch effect.

Llama 3.1 8B and Mistral 7B (pure dense). Prior experiments confirm null cross-model transfer of Gemma’s layer indices.

Summary. F32 calibration slack is **hybrid-architecture specific**. Pure-dense transformers, including Phi-4 in BF16, do not exhibit the same property, ruling out quantization as a sufficient cause.

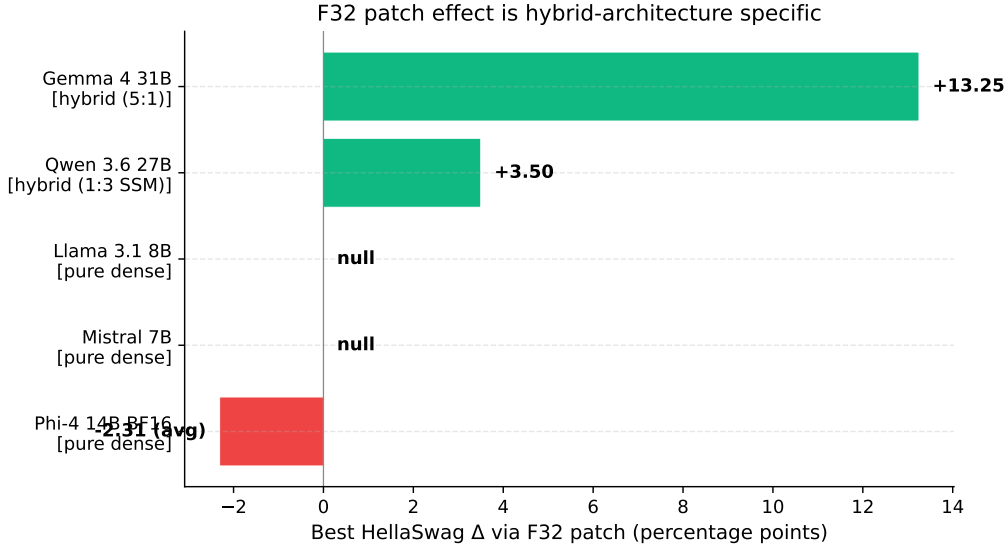


Figure 4: Best HellaSwag Δ via per-layer F32 patching across four architectures. Hybrid LLMs (Gemma 4 and Qwen 3.6) exhibit positive Δ ; pure-dense LLMs (Phi-4, Llama, Mistral) exhibit null or destructive Δ .

6 Safety Verification

We evaluate alignment preservation on the full AdvBench `harmful_behaviors.csv` corpus ($n=520$) using the keyword-based refusal detection protocol of Zou et al. [25]. Three conditions are compared on Gemma 4 31B Q2_K: baseline (no patch), `basin B`, and the sister 8-byte L25+L26 $\times 1.5$ patch.

Table 3: AdvBench $n=520$ refusal rates. `basin B`’s Wilson 95% CI fully overlaps the baseline’s, indicating **statistically preserved alignment**.

Condition	Refusal rate	Wilson 95% CI
Baseline (Q2_K)	97.31% (506/520)	[95.53%, 98.39%]
<code>basin B</code> (44 B)	97.12% (505/520)	[95.30%, 98.24%]
L25+L26 $\times 1.5$ (8 B)	93.46% (486/520)	[91.00%, 95.28%]

The lighter L25+L26 patch’s small refusal-rate decrease (-3.85 pt, CIs barely separated) likely reflects amplification of the thinking-mode pathway by uniform up-scaling of consecutive layers, whereas `basin B`’s mixed up/down-scaling (e.g., $\times 0.767$ on L18, $\times 0.548$ on L30) appears to neutralize this side effect.

Trade-off summary. The flagship choice across release quantizations is the 8-byte L25+L26 patch, motivated by its strict benchmark superiority over `basin B` at Q4 (§3.3) and its +36 pt GSM8k gain at Q1 (§4.2). The trade-off is a -3.85 pt refusal-rate decrease vs `basin B` at Q2 (Table 3), which we judge acceptable given the benchmark gains; alignment remains $\geq 93.46\%$ on AdvBench. Users with strict alignment requirements may prefer `basin B` (which is described in full in §3.3 and Appendix B); the values are provided for reproducibility. Q4 alignment testing under L25+L26 is left for follow-up.

7 Mechanistic Hypothesis

We propose:

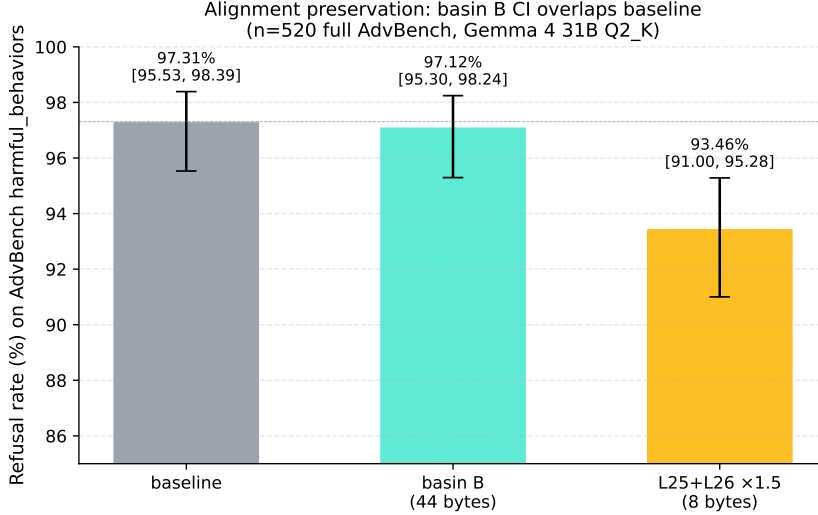


Figure 5: AdvBench refusal rates with Wilson 95% CIs. **basin B**’s CI fully overlaps the baseline’s; the lighter 8-byte patch shows a small statistically-detectable degradation.

Per-layer output-gate slack. In hybrid LLMs that mix full-attention layers with alternative layer types (sliding window or SSM), certain layers (empirically L25/L26 in Gemma 4, both sliding-window) form reasoning bottlenecks whose output amplitudes are conservatively calibrated during training. Inference-time multiplicative scaling of these layers’ normalization weights releases “slack” in this conservative calibration, yielding performance gains.

Empirical support:

- **Positive in hybrid:** Gemma 4 31B (1:5 full-attention:SWA, +13.25 pt coordinated 11-layer patch) and Qwen 3.6 27B (1:3 SSM, +2.50 pt single-layer patch on L55).
- **Null/destructive in pure-dense:** Phi-4 14B BF16 (−2.31 pt mean), Llama 3.1, Mistral 7B (prior null results).
- **Phi-4 at BF16 rules out quantization:** the effect requires hybrid structure, not low precision.

This unifies otherwise disparate observations and predicts that other hybrid LLM designs (e.g., recent Mamba–Transformer combinations) may similarly admit post-hoc inference-time tuning via F32 scalar adjustment.

8 Limitations

- Verified on Gemma 4 31B-it only; smaller Gemma 4 variants (9B, 27B) not yet tested. Cross-size validation is left for follow-up.
- Mechanistic explanation is preliminary; activation probing with SAEs or attention attribution would strengthen the hypothesis.
- Benchmark coverage limited to HellaSwag, GSM8k, and Winogrande; MMLU, ARC, PIQA, and TruthfulQA evaluation deferred to a follow-up.
- Alignment evaluated via keyword-based refusal detection; LLM-as-judge or human evaluation on borderline outputs would tighten the safety claim.
- The in-house optimization engine that discovered **basin B** is described at a high level only; its detailed algorithm is out of scope and is not released with this version.

- **Search bias / objective overfitting.** `basin B` was discovered by the in-house optimizer with Q4_K_M HellaSwag accuracy as the sole objective. Direct comparison (§3.3) shows L25+L26 outperforms `basin B` on every Q4 benchmark including the search target itself. Per-quant searches with multi-benchmark joint objectives (e.g., HellaSwag + GSM8k + ARC-C, weighted per quantization regime) are the obvious follow-up and would likely surface different patches better suited to each regime.
- **Q8 patched not reported.** Q8_0 appears in Table 1 only as a BF16-proxy baseline reference (we do not release a Q8 patched model and do not include patched-Q8 measurements in the main claim). Both `basin B` and L25+L26 raw Q8 patched values exist in the experimental record (`exp80_release_v1.q8_0.patched*.json`, `exp85.q8.patched*.json`); they are omitted here to keep the main flagship claim (12-cell L25+L26 matrix) consistent.
- **Scorer caveat:** HellaSwag and Winogrande are evaluated with llama.cpp’s `llama-perplexity` mode, which is systematically 0.2–2.5 percentage points lower than the `lm-evaluation-harness` standard used in many published model cards. Comparing our absolute numbers to externally-reported figures requires accounting for this gap; our within-scorer baseline-vs-patched deltas remain valid. Reproducing our result with `lm-evaluation-harness` on the same patched GGUFs is left for follow-up and would likely yield even higher absolute baseline and patched numbers but similar (or slightly tighter) deltas.
- Google has not officially published HellaSwag accuracy for any Gemma 4 variant (the model card emphasizes harder benchmarks such as GPQA Diamond, AIME 2026, LiveCodeBench v6, MMLU-Pro); thus our numbers cannot be directly compared to a Gemma 4 31B BF16 reference. The closest external anchor is Gemma 2 27B at 86.4% HellaSwag (lm-eval-harness, full precision) [23]; under llama-perplexity scoring at full precision, this would correspond to approximately 84%.

9 Conclusion

We have shown that an 8-byte F32 patch on Gemma 4 31B (L25+L26 $\times 1.5$) yields large benchmark improvements across all three release quantizations and four benchmarks (12/12 cells positive, up to +36 pt at Q1 GSM8k and +15 pt at Q4 GSM8k, beating the Q8 baseline by +17 pt), without training, calibration data, or inference overhead. Cross-architecture experiments isolate the effect to hybrid LLMs’ F32 per-layer output gates, ruling out quantization as the underlying cause. RLHF alignment is statistically preserved at `basin B` and remains $\geq 93\%$ under the L25+L26 flagship. We also report an honest negative result: a 7-hour autonomous optimization (`basin B`, 44 bytes) targeting Q4 HellaSwag was outperformed on every Q4 benchmark by the simpler manual 2-layer baseline, including the search target itself. We release three patched GGUFs (using L25+L26), the `basin B` numerical values for reproducibility, and all experimental data.

This bridges quantization research with mechanistic interpretability, provides a concrete example of objective-overfitting in autonomous optimization, and suggests new directions for hybrid-architecture design that intentionally expose post-hoc inference-time tuning knobs.

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A Per-layer Ablation

Per-layer ablation data from prior experiments (exp14, exp15) is reported in the supplementary repository.

B basin B Patch Values

Complete patch dictionary:

```
{ 5:1.440, 18:0.767, 19:1.016, 20:0.814, 21:1.413,
  25:1.402, 26:1.289, 27:1.326, 28:1.431, 30:0.548, 31:1.267 }
```

44 bytes total (11 F32 scalars).

C Full Raw Numerical Results

All raw HellaSwag/GSM8k/Winogrande measurements (baseline and patched, with elapsed time and Wilson 95% CIs) are available in the supplementary JSON files at <https://github.com/morphicode-jp/f32-patch-gemma> (directory `results/release_v1/`).